Heston model: Andersen's Quadratic-Exponential (QE) scheme

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May 2026; revised October 2026

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1 Introduction and goals

Full-truncation Euler (see the note *Heston model: Monte Carlo simulation with full-truncation Euler*) is the baseline scheme for Heston Monte Carlo. The scheme is robust, simple, and works correctly across all parameter regimes, but it carries a discretisation bias of order $O(\Delta t)$ that becomes problematic in two specific situations:

- Low Feller parameter $\nu = 2\kappa\theta/\sigma^2$, where the variance process spends measurable time near $v = 0$ and the truncation $v_n^+ = \max(\hat{v}_n, 0)$ fires frequently. The asymmetric distribution of v_t near zero is poorly captured by a Gaussian Euler step.
- Short maturities relative to the discretisation step, where the bias dominates the statistical error and large N are needed for any pretense of accuracy.

The remedy is to exploit a property the Heston model gives us *for free*: the conditional distribution of $v_{t+\Delta} \mid v_t$ is known explicitly, in closed form, as a non-central chi-squared (the Heston model note, §3.3). A simulator that samples $v_{t+\Delta}$ from this exact distribution would be bias-free *within the variance process*, with discretisation error coming only from the spot equation. Such a sampler exists and was constructed by Broadie and Kaya (2006); we discuss it below.

The Broadie-Kaya method is bias-free but expensive: each step requires sampling a non-central chi-squared variate, which involves an iterative algorithm (typically rejection sampling or numerical inversion). Andersen (2008) observed that for the dynamics of a Heston call option, an *approximate* simulator that matches the first two conditional moments of $v_{t+\Delta} \mid v_t$ is sufficient in practice: the bias left by the unmatched higher moments is small at typical step sizes, as his numerical experiments show. Andersen's Quadratic-Exponential (QE) scheme is the workhorse of this insight.

This note has four goals.

Theoretical foundation. We derive the conditional moments of the CIR process explicitly, and use them to motivate the moment-matching strategy.

The QE scheme. We construct Andersen's quadratic regime (for high v_t) and exponential regime (for low v_t), and the discriminator ψ that switches between them.

The log-spot integration. Once v_t and $v_{t+\Delta}$ are sampled, the spot SDE benefits from a refined integration that uses both endpoints; we derive this central discretisation.

Validation. We confirm experimentally that QE has substantially lower bias than full-truncation Euler in the difficult regimes (low ν, short maturities), and we use the Fourier pricer as ground truth.

2 Why full-truncation Euler is not enough

2.1 The bias mechanism

As noted in the full-truncation Euler note, the truncation $v_n^+ = \max(\hat{v}_n, 0)$ fires with non-negligible probability when $\hat{v}_n$ is small. Each truncation event contributes a small bias: rather than letting

the "true" continuous variance reach zero and immediately reflect upward (the regime captured by the non-central chi-squared distribution), the discretised process is artificially pinned at zero for one step, missing both the negative diffusive impulse and the structural drift back towards θ.

Lord, Koekkoek and van Dijk (2010) compare the Euler-type fixes for negative variance and find full truncation the least biased of them; its weak error is still first order in Δt, with a constant that grows with the time the process spends near zero. For $\nu \ll 1$ the constant is large. In this repository's aggressive case ($\nu = 0.04$, $T = 1$, `benchmark_phase4.py`, Section 5) the bias of full-truncation Euler is about 0.8% of the option price at $\Delta t = 0.005$, too large for a calibration that needs 0.1% accuracy.

2.2 The right diagnostic

To diagnose the issue empirically, the right experiment is: fix M at a large value (so statistical error is negligible), vary N, and plot the bias $|\widehat{C}(N) - C_{\text{Fourier}}|$ on a log-log scale. For full-truncation Euler in the low-ν regime, the slope is approximately -1 (weak rate 1), but the absolute error at typical N is large enough to motivate a better scheme.

For Andersen QE in the same regime the bias is much smaller at every N; in this repository's study it stays below the Monte Carlo resolution from $N = 25$ upwards, so its slope cannot be measured at these sample sizes (see the empirical comparison below).

3 Exact simulation: Broadie-Kaya (conceptual)

For completeness we describe the Broadie-Kaya (2006) exact simulation scheme, even though we do not implement it. The reader can think of QE as a numerical approximation of this exact algorithm that trades a small amount of accuracy for a large saving in cost.

3.1 The Broadie-Kaya construction

The construction is built on three exact distributional facts about the Heston model:

1. **Variance transition.** As established in the Heston model note, §3.3, the conditional distribution of $v_{t+\Delta} \mid v_t$ is non-central chi-squared, scaled. Specifically, if we let

$$d = \frac{4\kappa\theta}{\sigma^2}, \qquad c(\Delta) = \frac{\sigma^2(1 - e^{-\kappa\Delta})}{4\kappa},$$

   then

$$v_{t+\Delta} \;\sim\; c(\Delta) \cdot \chi^2_d\!\left(\frac{v_t e^{-\kappa\Delta}}{c(\Delta)}\right),$$

   where $\chi^2_d(\lambda)$ denotes a non-central chi-squared with d degrees of freedom and non-centrality λ.

2. **Integrated variance.** Conditional on v_t and $v_{t+\Delta}$, the integral $\int_t^{t+\Delta} v_s\, ds$ has a known characteristic function (Pitman-Yor 1982), which can be inverted numerically.

3. **Spot increment.** Conditional on the variance path, the log-spot increment is Gaussian with mean and variance computable from the integrated variance.

The Broadie-Kaya algorithm is to sample $v_{t+\Delta}$ from the non-central chi-squared, then sample the integrated variance from its conditional distribution (numerical inversion of the characteristic function), then sample the conditional Gaussian for the log-spot.

The result is bias-free in distribution: every step is a draw from the exact transition kernel of the Heston SDE, with no truncation, no Euler approximation, no time-discretisation residual.

3.2 Why we don't use it

The cost is dominated by step 2: sampling the integrated variance from a numerically-inverted characteristic function, which requires many evaluations of that characteristic function for every draw, with a count that grows with the precision target. Compare this to QE's per-step cost of $O(1)$ operations: a uniform draw, two normal draws, and a few arithmetic operations. Broadie-Kaya is therefore much more expensive per step than QE; we have not implemented it, so this repository has no measurement of the factor.

In production trading, this cost gap matters. Andersen's insight is that if we are willing to accept a small residual bias instead of zero, we can recover almost all of the precision benefit at a fraction of the cost.

4 The Andersen QE scheme: conditional moments

The starting point of QE is the explicit form of the first two conditional moments of $v_{t+\Delta} \mid v_t$ under CIR dynamics. These are known in closed form and form the basis of the moment-matching approximation.

4.1 Conditional mean

For the CIR process $dv_t = \kappa(\theta - v_t)dt + \sigma\sqrt{v_t}\, dW_t$, integration of the deterministic part gives

$$m(v_t, \Delta) := \mathbb{E}[v_{t+\Delta} \mid v_t] = \theta + (v_t - \theta)\, e^{-\kappa\Delta}. \tag{1}$$

This is the analytic mean-reversion: v relaxes towards θ at exponential rate κ. As $\Delta \to 0$, $m \to v_t$; as $\Delta \to \infty$, $m \to \theta$.

4.2 Conditional variance

For the variance, the calculation requires Itô's lemma applied to v^2:

$$d(v^2) = 2v\, dv + d\langle v\rangle = (2v\,\kappa(\theta - v) + \sigma^2 v)\, dt + 2v\sigma\sqrt{v}\, dW.$$

Taking expectations and integrating $\mathbb{E}[v_s^2]$ as an ODE in s, then computing $\mathrm{Var}(v_{t+\Delta} \mid v_t) = \mathbb{E}[v_{t+\Delta}^2 \mid v_t] - m^2$, one obtains

$$s^2(v_t, \Delta) := \mathrm{Var}(v_{t+\Delta} \mid v_t) = v_t\, \frac{\sigma^2 e^{-\kappa\Delta}}{\kappa}\,(1 - e^{-\kappa\Delta}) + \theta\, \frac{\sigma^2}{2\kappa}\,(1 - e^{-\kappa\Delta})^2. \tag{2}$$

The first term is proportional to v_t (it dominates when v_t is large); the second is independent of v_t (it dominates when v_t is small). Both vanish as $\Delta \to 0$ as expected, and they tend to a finite limit (the stationary variance of CIR) as $\Delta \to \infty$.

4.3 The discriminator ψ

The dimensionless ratio

$$\psi(v_t, \Delta) := \frac{s^2(v_t, \Delta)}{m(v_t, \Delta)^2} \tag{3}$$

is the central object of Andersen's construction. It measures the relative dispersion of the conditional distribution: small ψ means $v_{t+\Delta}$ is concentrated around its mean (a Gaussian-like shape works

well), while large ψ means the distribution is dispersed, with much of its mass close to zero (a Gaussian fits poorly).

For Heston with typical parameters and $\Delta = 0.01$:

- When $v_t \sim \theta$ (around the long-run mean), ψ is small: $\psi \approx \sigma^2\Delta/\theta$ to leading order in Δ, which gives 0.022 for the standard set ($\kappa = 1.5$, $\sigma = 0.3$) and 0.25 for the aggressive set ($\kappa = 0.5$, $\sigma = 1$) of the validation section.
- As v_t decreases, ψ increases (the derivative of s^2/m^2 in v_t is negative for $v_t > 0$), up to its maximum at $v_t = 0$, computed in the next bullet. For $\nu < 1$ the boundary at zero is attainable and the conditional density piles up near zero, which the exponential regime approximates by an atom.
- In the boundary case $v_t = 0$ exactly, $m = \theta(1 - e^{-\kappa\Delta})$ and $s^2 = (\sigma^2\theta/2\kappa)(1 - e^{-\kappa\Delta})^2$, so $\psi = (\sigma^2)/(2\kappa\theta) = 1/\nu$, for every Δ. Hence $\psi \leq 1/\nu$ everywhere, and the exponential regime ($\psi > \psi_c = 1.5$) is entered only when $\nu < 2/3$: never for the standard set ($\psi \leq 0.75$), and for the aggressive set ($1/\nu = 25$) whenever v_t is small. The smaller the Feller parameter, the larger ψ at the boundary, the more important QE's atom-at-zero treatment.

The QE scheme uses ψ to decide between two distinct sampling strategies, glued at a critical threshold ψ_c that Andersen recommends to be $\psi_c = 1.5$ based on numerical experiments.

5 The two regimes of QE

5.1 Quadratic regime ($\psi \leq \psi_c$)

When the conditional distribution is reasonably concentrated (small ψ), Andersen approximates it by the distribution of $a(b + Z)^2$ where $Z \sim \mathcal{N}(0, 1)$. The unknowns a, b are two free parameters fixed by matching the first two moments:

$$\mathbb{E}[a(b + Z)^2] = a(1 + b^2), \tag{4}$$

$$\mathrm{Var}(a(b + Z)^2) = 2a^2(1 + 2b^2). \tag{5}$$

Setting (4) $= m$ and (5) $= s^2$:

$$\frac{s^2}{m^2} = \psi = \frac{2(1 + 2b^2)}{(1 + b^2)^2}.$$

Solving for b^2:

$$b^2 = 2\psi^{-1} - 1 + \sqrt{2\psi^{-1}(2\psi^{-1} - 1)}. \tag{6}$$

This requires $2\psi^{-1} \geq 1$, i.e., $\psi \leq 2$. For $\psi \in (0, 2]$ this gives $b^2 \geq 0$, which is well-defined; the upper bound $\psi = 2$ corresponds to $b^2 = 0$ (a centered chi-squared, all mass on the right half-line). Once b^2 is determined, a follows:

$$a = \frac{m}{1 + b^2}. \tag{7}$$

The sampling step is straightforward: draw $Z \sim \mathcal{N}(0, 1)$, set $\hat{v}_{n+1} = a(b + Z)^2$. This is non-negative by construction.

Properties of the quadratic regime. The samples are non-negative (good), the first two moments match exactly (good), but the higher moments do not match. The third moment, in particular, is captured only approximately, leaving a small residual bias in the option price; this is what we trade for the cheap sampling.

5.2 Exponential regime ($\psi > \psi_c$)

When the conditional distribution is highly dispersed (large ψ), Andersen approximates it by a mixture: with probability p the variance is exactly zero, and with probability $1-p$ it is exponentially distributed with rate β. The CDF of this mixture is

$$F(v) = \begin{cases} p + (1-p)(1-e^{-\beta v}), & v \geq 0 \\ 0, & v < 0. \end{cases} \tag{8}$$

The two parameters p, β are again fixed by matching the first two moments. Computing:

$$\begin{aligned} \mathbb{E}[v_{t+\Delta}] &= (1-p)/\beta, \\ \mathrm{Var}(v_{t+\Delta}) &= (1-p)/\beta^2 + p(1-p)/\beta^2 = (1-p)(1+p)/\beta^2. \end{aligned}$$

Setting these equal to m and s^2 and dividing:

$$\frac{s^2}{m^2} = \psi = \frac{(1-p)(1+p)/\beta^2}{(1-p)^2/\beta^2} = \frac{1+p}{1-p}.$$

Solving for p and β:

$$p = \frac{\psi - 1}{\psi + 1}, \qquad \beta = \frac{2}{m(\psi+1)} = \frac{1-p}{m}. \tag{9}$$

For $\psi > 1$, $p \in (0,1)$ and the mixture is well-defined. (For $\psi = 1$, $p = 0$ and the distribution is purely exponential with $\beta = 1/m$; for $\psi \to \infty$, $p \to 1$ and the distribution becomes purely an atom at zero.)

The sampling step uses inverse-CDF: draw $U \sim \mathrm{Uniform}(0,1)$, and

$$\hat{v}_{n+1} = \begin{cases} 0, & U \leq p, \\ -\frac{1}{\beta}\log\left(\frac{1-U}{1-p}\right), & U > p. \end{cases} \tag{10}$$

The atom at zero stands in for the mass that the exact transition law puts close to zero (the exact law has a density and no atom); this is essential for low-ν regimes.

Why two regimes. The natural question is: why not use the exponential regime everywhere? The answer is that for small ψ, the moment-matching equations (9) give $p < 0$ (no mixture interpretation possible), so the exponential regime is not even well-defined. Conversely, the quadratic regime requires $\psi \leq 2$. The threshold $\psi_c \in (1,2)$ is a free parameter of the scheme; Andersen recommends $\psi_c = 1.5$ based on extensive empirical testing.

5.3 Continuity at the threshold

A subtle but important fact: at $\psi = \psi_c = 1.5$, both regimes give well-defined (though different) approximating distributions. The two are continuous in their first two moments by construction, but *the higher moments* jump at the threshold. In practice this discontinuity is invisible because:

- Each path uses one regime per step, not both.
- Aggregating across paths and steps, the threshold is crossed many times in opposite directions and the third-moment jumps average out to a small bias.

The lack of higher-moment continuity means that QE is not a smooth interpolation between two limiting samplers; it is two distinct samplers spliced at a parameter-dependent threshold. This is fine for option pricing where we care about $\mathbb{E}[f(S_T)]$, but would be problematic for path-dependent quantities like maxima or local times.

6 The log-spot scheme

Once v_t and $v_{t+\Delta}$ are sampled (by either regime), we still need to discretise the log-spot SDE. Recall from the full-truncation Euler note (equation (6) there) that

$$d\log S_t = (r - \tfrac{1}{2}v_t)\,dt + \sqrt{v_t}\,dW_t^1,$$

where W^1 has correlation ρ with the variance Brownian motion W^2. Full-truncation Euler uses a simple Euler step

$$\log \hat{S}_{n+1} - \log \hat{S}_n = (r - \tfrac{1}{2}v_n^+)\,\Delta t + \sqrt{v_n^+}\,\Delta W_n^1,$$

which uses v_n (the left endpoint) for both the drift and the diffusion. Since QE samples the right endpoint v_{n+1} explicitly, we can do better.

6.1 Splitting the Brownian motion

The key tool is the Cholesky decomposition of the joint Brownian motion: if W^1 and W^2 have correlation ρ, then we can write

$$dW_t^1 = \rho\,dW_t^2 + \sqrt{1-\rho^2}\,dW_t^\perp, \tag{11}$$

where $W^\perp$ is a Brownian motion independent of W^2. Substituting:

$$d\log S_t = (r - \tfrac{1}{2}v_t)\,dt + \rho\sqrt{v_t}\,dW_t^2 + \sqrt{(1-\rho^2)v_t}\,dW_t^\perp. \tag{12}$$

The second term, $\rho\sqrt{v_t}\,dW_t^2$, can be eliminated using the variance SDE: from $dv_t = \kappa(\theta - v_t)dt + \sigma\sqrt{v_t}\,dW_t^2$, we have $\sqrt{v_t}\,dW_t^2 = (1/\sigma)(dv_t - \kappa(\theta - v_t)dt)$. Substituting:

$$d\log S_t = \left(r - \tfrac{1}{2}v_t - \tfrac{\rho\kappa\theta}{\sigma} + \tfrac{\rho\kappa}{\sigma}v_t\right)dt + \tfrac{\rho}{\sigma}\,dv_t + \sqrt{(1-\rho^2)v_t}\,dW_t^\perp. \tag{13}$$

The dv_t on the right is exactly the variance increment, which we know after sampling: $\Delta v = \hat{v}_{n+1} - \hat{v}_n$. The $dW^\perp$ is independent of everything we have already sampled. So integrating (13) over a step gives an expression where every term except the last can be evaluated from quantities we already know.

6.2 The Andersen central discretisation

Andersen parameterises the integration of (13) with two free constants $\gamma_1, \gamma_2 \geq 0$ satisfying $\gamma_1 + \gamma_2 = 1$, which weight the contributions of v_n and v_{n+1} to the integrals over the step. The resulting update is

$$\log \hat{S}_{n+1} = \log \hat{S}_n + K_0 + K_1\hat{v}_n + K_2\hat{v}_{n+1} + \sqrt{K_3\hat{v}_n + K_4\hat{v}_{n+1}}\,Z, \tag{14}$$

where $Z \sim \mathcal{N}(0,1)$ is independent of all variance increments, and the coefficients are:

$$K_0 = \left(r - \tfrac{\rho\kappa\theta}{\sigma}\right)\Delta t, \tag{15}$$

$$K_1 = \gamma_1\Delta t\left(\tfrac{\kappa\rho}{\sigma} - \tfrac{1}{2}\right) - \tfrac{\rho}{\sigma}, \tag{16}$$

$$K_2 = \gamma_2\Delta t\left(\tfrac{\kappa\rho}{\sigma} - \tfrac{1}{2}\right) + \tfrac{\rho}{\sigma}, \tag{17}$$

$$K_3 = \gamma_1\Delta t(1-\rho^2), \tag{18}$$

$$K_4 = \gamma_2\Delta t(1-\rho^2). \tag{19}$$

The natural choice is $\gamma_1 = \gamma_2 = 1/2$ (Crank-Nicolson-like centering of the integration), which we adopt. With these values, the scheme is symmetric in $\hat{v}_n, \hat{v}_{n+1}$ and matches the conditional log-spot variance to higher order in Δt than the Euler version.

Why this matters. The advantage of (14) over a simple Euler is that the integrand in the $dW^{\perp}$ noise is now $\sqrt{K_3\hat{v}_n + K_4\hat{v}_{n+1}}$ rather than $\sqrt{v_n^+}$. The former is the trapezoidal approximation of $\int_t^{t+\Delta} v_s\, ds$ and is more accurate; the latter is the left-endpoint approximation. This is the same idea as Crank-Nicolson over Implicit Euler in PDE land.

7 Convergence and bias

We discuss the weak convergence properties of QE and contrast with full-truncation Euler.

7.1 Theoretical results

Andersen's paper does not prove a weak convergence rate for QE; it gives extensive numerical evidence that its bias at practical step sizes is far smaller than that of Euler-type schemes, above all when the Feller condition fails. Full-truncation Euler has first-order weak error. We make no claim about the asymptotic rate of QE: the comparison below is empirical.

The reason for the smaller bias is visible in the construction:

- Both schemes match the first conditional moment of v exactly.
- Only QE matches the second conditional moment exactly. Full-truncation Euler matches it up to $O(\Delta t^2)$ when no truncation fires, but has $O(\Delta t)$ error when truncation does fire.
- QE reproduces the concentration of v near zero through the atom of its exponential regime, which Euler's truncation can only imitate step by step.

7.2 Empirical comparison

The bias study in `benchmark_phase4.py` (Section 5) uses the aggressive parameter set $\kappa = 0.5$, $\theta = 0.04$, $\sigma = 1$, $\rho = -0.9$, $v_0 = 0.04$ ($\nu = 0.04$), $S_0 = K = 100$, $r = 0.03$, $T = 1$, with 10^5 paths per run averaged over three seeds and the Fourier price 6.7304 as reference (`results/phase4/heston_qe_vs_euler.csv`):

N	FT-Euler |bias|	QE |bias|
25	0.688	0.007
50	0.293	0.003
100	0.119	0.002
200	0.056	0.007
400	0.032	0.011

The 95% half-width of the three-seed average is about 0.018 (some 0.26% of the price; column `qe_hw_avg` of the CSV), so every QE value is within one half-width of zero: no QE bias is resolved at these sample sizes. Euler's bias falls roughly as $1/N$, as first-order weak convergence predicts, while QE at $N = 25$ is already more accurate than Euler at $N = 400$. The validator repeats the comparison at $r = 0.05$, $T = 0.5$, $N = 50$ with five seeds of 2×10^5 paths: Euler's bias is $+0.155$, QE's -0.004, within noise.

With the canonical parameters of `benchmark_phase4.py` ($\kappa = 2$, $\theta = 0.04$, $\sigma = 0.3$, $\rho = -0.7$, $v_0 = 0.04$, so $\nu \approx 1.78$; $r = 0.03$, $T = 1$) the variance process rarely approaches zero and the two schemes agree with the Fourier price within their confidence intervals at $N = 200$ for all nine strikes from 80 to 120 (`results/phase4/heston_cross_validation.csv`); the advantage of QE is specific to the low-ν regime.

7.3 Cost considerations

QE has a higher per-step cost than FT-Euler:

- FT-Euler: 2 normals per step, plus arithmetic.
- QE: 1 uniform + 2 normals per step (the uniform U is used by the exponential regime, the normal Z by the quadratic regime, and the second normal drives the log-spot; all three are drawn at every step), plus arithmetic for the moment computation and ψ check.

In the Python implementation a QE step costs about 1.5× a full-truncation Euler step (timings in `results/phase4/heston_cross_validation.csv`). In the aggressive regime this is easily repaid, since QE at $N = 25$ beats Euler at $N = 400$; in the benign regime the two are equally accurate at the same N and Euler is the cheaper choice.

8 Validation strategy

The validation strategy mirrors that of full-truncation Euler, but with stricter tolerances reflecting QE's higher accuracy.

8.1 Pillars

The pillars below are the tests of `validate_heston_qe.py`.

1. BS limit. As $\sigma \to 0$, QE must reduce to GBM and the call price must converge to BS. This is the cheapest sanity check.

2. Cross-method consistency. For the standard and the aggressive parameter sets and $T \in \{0.25, 0.5, 1\}$, QE estimates must agree with the Fourier reference within statistical half-widths.

3. Statistical convergence. At fixed N, the half-width times $\sqrt{M}$ must be approximately constant in M.

4. Antithetic variates. The antithetic version must reduce the variance by at least a factor of two and agree with the plain estimator.

5. Bias comparison vs FT-Euler. In the aggressive regime at $N = 50$, QE's bias must be smaller than full-truncation Euler's by at least a factor of five.

8.2 Numerical reference

For the standard Heston parameters ($\kappa = 1.5, \theta = 0.04, \sigma = 0.3, \rho = -0.7, v_0 = 0.04$), $S_0 = K = 100$, $r = 0.05$:

T	Fourier	Comments
0.25	4.5842062306	Short maturity, small bias
0.5	6.8257341234	Standard test case
1.0	10.3618690210	Longer maturity

9 Implementation outline

9.1 Python

The module `quantlib.heston_qe` exposes:

- `simulate_terminal_heston_qe(S0, v0, r, kappa, theta, sigma, rho, T, n_steps, n_paths, *, seed=None, rng=None, antithetic=False, psi_c=1.5, gamma1=0.5, gamma2=0.5)`: terminal-only QE simulator with $O(M)$ memory, paralleling `simulate_terminal_heston` from `quantlib.heston_mc`.
- `mc_european_call_heston_qe(...)`: high-level pricer with the same interface as `mc_european_call_heston`

A full-grid simulator `simulate_heston_paths_qe` is not provided; path-dependent QE pricing lives in `quantlib.heston_exotics`.

9.2 C++

The C++ implementation in `cpp/include/quant/heston_qe.hpp` and `cpp/src/heston_qe.cpp` mirrors the Python interface using `HestonParams` from `heston_fourier.hpp` and reusing `MCResult` from `monte_carlo.hpp`. The internal helper `qe_step` takes the same role as `ft_step` in `heston_mc.cpp`: a single function that captures the entire scheme update, callable from both the path and terminal simulators.

The Catch2 tests in `cpp/tests/test_heston_qe.cpp` verify the four pillars at reduced sample size and snapshot against Python references, mirroring the structure of `test_heston_mc.cpp`.

10 Bibliographical notes

- Andersen (2008) [1]: the seminal QE paper, with detailed parameter tables and benchmarks. The reference for this entire note; readable and well-organized.
- Broadie, Kaya (2006) [2]: the exact simulation algorithm. Worth reading once to understand what QE is approximating.
- Lord, Koekkoek, van Dijk (2010) [4]: comparison of the Euler-type fixes for negative variance; full truncation has the smallest bias among them.
- Smith (2007) [5]: an almost exact simulation method that extends Broadie–Kaya to payoffs monitored at many dates.
- Glasserman, Kim (2011) [3]: an alternative exact simulation by gamma expansion; cheaper than Broadie-Kaya but still more expensive than QE.

References

[1] Andersen, L. (2008). *Simple and efficient simulation of the Heston stochastic volatility model.* Journal of Computational Finance, 11(3), 1–42.

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